

INPUT

Travel demand

Origin-destination flows
of freight goods (tonnes)

Graph technology

Nodes (NUTS-2 regions) | edges
(corridor connections) | route
definitions

Parameters on charging infrastructure expansion

Maximum possible speed of build-up

Costs of drivetrain technologies

CAPEX & OPEX per technology

BEV technology

Battery capacity | charging power
| specific energy consumption

Mandatory break regulations

EU driving time rules | minimum
cumulative travel times at break
locations

Geographically varying charging costs

Market-zone electricity-prices | location-
specific grid charges

Modal shift to rail

Maximum rail share | rate-of-change
constraints | Terminal and variable costs
of rail transport

Optimization model

- **Linear program**
- **based on Network Design Problem**
- System cost-minimization
- Geographic resolution: NUTS-2
- Temporal resolution: annual
- Network-based transport demand coverage
- Technology & mode shift
- Sizing and allocation of charging infrastructure

OUTPUT

Freight flow per drivetrain
technology

Vehicle stock shift

Charging load by location

Charging infrastructure expansion

Mode split
(Road vs. rail)